

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **LOKESH S (192572151)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

LOKESH S

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ →
Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

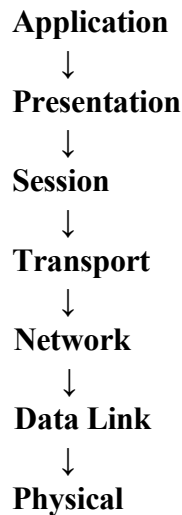
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

1. Missing OSI Layers and Their Functions

Completed Concept Map



Missing Layers

- Session Layer
- Data Link Layer

Analysis

Session Layer

- Establishes communication between devices.
- Maintains the communication session.
- Synchronizes data exchange.
- Terminates the session after communication.

Importance

- Prevents communication interruptions.
- Allows interrupted transfers to resume.
- Maintains proper synchronization between sender and receiver.

Example

During a video meeting, if the internet disconnects briefly, the session layer helps maintain or recover the communication session.

Data Link Layer

- Transfers data between directly connected devices.
- Converts packets into frames.
- Performs error detection using CRC.
- Controls access to the physical medium using MAC addresses.

Importance

- Detects transmission errors.
- Ensures reliable local communication.
- Prevents corrupted frames from reaching higher layers.

Example

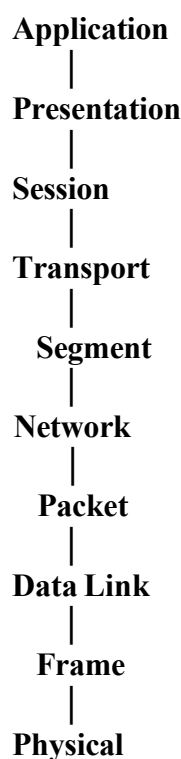
When a laptop sends data to a Wi-Fi router, the Data Link layer uses MAC addresses for local delivery.

Conclusion

The Session Layer manages communication sessions, while the Data Link Layer provides reliable local transmission. Without these layers, communication would become unstable and error-prone.

2. Concept Map for Data Units in the OSI Model

Concept Map



|
Bits

Analysis

OSI Layer	Data Unit	Function
Transport	Segment	Divides large data into manageable pieces.
Network	Packet	Adds logical (IP) addressing and routing information.
Data Link	Frame	Adds MAC addresses and error detection.
Physical	Bits	Converts data into electrical, optical, or radio signals.

Transformation Process

1. Application Layer
 - Creates the original message.
2. Transport Layer
 - Breaks data into segments.
3. Network Layer
 - Adds IP information to form packets.
4. Data Link Layer
 - Adds MAC address and CRC to create frames.
5. Physical Layer
 - Converts frames into bits for transmission.

Example

Sending an email:

Email

↓

Segment

↓

Packet

↓

Frame

↓

Bits

Each transformation adds the information needed for reliable communication.

Conclusion

Data transformation allows each OSI layer to perform its specific task, ensuring efficient and accurate transmission from sender to receiver.

3. MAC Address vs IP Address Concept Map

Concept Map

Data Link Layer



MAC Address



Local Delivery

Network Layer



IP Address



Global Delivery

Analysis

MAC Address

- Physical address of a device.
- Used within the same local network (LAN).
- Fixed by the manufacturer.
- Operates at the Data Link layer.

Example

A computer sending data to a nearby printer.

IP Address

- Logical address assigned to a device.
- Used for communication between different networks.
- Can change depending on the network.
- Operates at the Network layer.

Example

Opening a website located in another country.

How They Work Together

Suppose a laptop accesses www.google.com.

1. IP address identifies Google's server.
2. Router forwards packets based on IP.
3. On each local network, MAC addresses deliver frames to the next device.
4. Process repeats until destination.

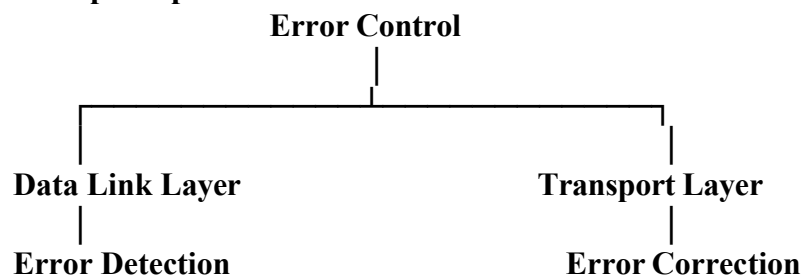
**Laptop
(MAC)**
↓
**Router
(MAC)**
↓
**Internet
(IP Routing)**
↓
**Google Server
(MAC)**

Conclusion

- IP Address provides global routing.
- MAC Address provides local delivery.
- Together they ensure complete communication across networks.

4. Error Control Concept Map

Concept Map



Analysis

Data Link Layer

Responsibilities:

- Detects transmission errors.
- Uses CRC and checksum.
- Discards damaged frames.

Transport Layer

Responsibilities:

- Ensures end-to-end reliability.
- Uses acknowledgments (ACKs).
- Retransmits lost segments.
- Maintains correct data order.

Interaction

Suppose 100 frames are transmitted.

- Data Link detects corrupted Frame 15.
- Receiver discards it.
- Transport notices missing data.
- Sender retransmits the lost segment.
- Complete message is reconstructed.

Benefits

- Accurate communication.
- Reduced data corruption.
- Reliable file transfers.
- Better network performance.

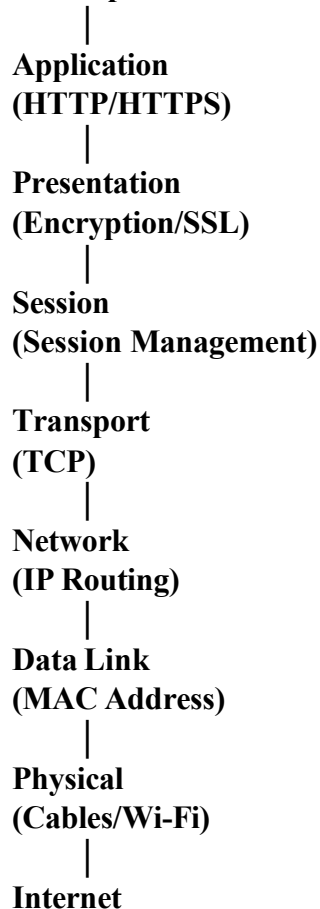
Conclusion

The Data Link layer detects errors on each link, while the Transport layer recovers from lost or corrupted data, providing reliable end-to-end communication.

5. Concept Map for Web Browsing Using the OSI Model

Concept Map

User Opens Website



Analysis

Layer	Role in Web Browsing
Application	Sends HTTP/HTTPS request.
Presentation	Encrypts and decrypts data using SSL/TLS.
Session	Establishes and maintains communication.
Transport	Ensures reliable TCP transmission.
Network	Routes packets to the web server.
Data Link	Delivers frames within the local network.
Physical	Transmits bits through cables or wireless signals.

Failure Analysis

Physical Layer Failure

- Cable disconnected.
- No internet connectivity.

Data Link Failure

- MAC communication fails.
- Frames cannot reach the router.

Network Failure

- Incorrect IP configuration.
- Website becomes unreachable.

Transport Failure

- TCP connection fails.
- Web page loads incompletely or times out.

Session Failure

- Session disconnects.
- User may be logged out unexpectedly.

Presentation Failure

- SSL/TLS issues.
- Browser displays a certificate or security error.

Application Failure

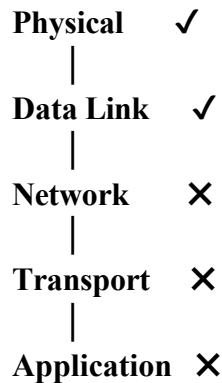
- Browser or web server malfunction.
- Website does not load.

Conclusion

Every OSI layer depends on the layers below it. A failure in any layer can disrupt the entire web browsing process.

6. Network Troubleshooting Using the Concept Map

Concept Map



Analysis

What the Symbols Mean

- ✓ = Layer is functioning correctly.
- ✗ = Layer has a problem.

Problem Identification

Since the Physical and Data Link layers are working:

- Cables or Wi-Fi are connected.
- Local communication is successful.

The Network Layer has failed:

- IP configuration may be incorrect.
- Routing problem.
- Gateway may be unreachable.
- Router may be malfunctioning.

Because the Network layer fails:

- Transport layer cannot establish TCP/UDP communication.
- Applications cannot send or receive data.
- Internet services stop working.

Troubleshooting Steps

1. Check IP address configuration.
2. Verify subnet mask and default gateway.
3. Test connectivity using ping to the gateway.
4. Check router status and routing table.
5. Confirm DNS settings if IP routing is working but names cannot be resolved.
6. Restart the network adapter or router if needed.

Conclusion

Concept maps clearly illustrate how OSI layers depend on one another. They make it easier to understand data flow, identify each layer's responsibilities, and troubleshoot network problems systematically, making them valuable tools for learning and maintaining computer networks.

